

Chandrayaan-2 Dual Frequency SAR User Manual



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Document Control & Data Sheet

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Contents

Document Control & Data Sheet	2
Document Change History	3
Acknowledgements.....	4
Contents.....	5
1. Objective.....	1
2. Data Products Archival	1
2.1. Archived Data Products.....	2
2.1.1 Raw data	2
2.1.2 Calibrated Data Products.....	3
3. Tools.....	6
4. Conclusion	6
5. Abbreviations.....	6
References	7

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1. Objective

The objective of this document is to provide basic overview about Chandrayaan-2 DFSAR data products. It brings out the basic organization of data , number of data products, types and formats available for the users to work with PDS4 data products.

2. Data Products Archival

Archive structure is designed both at mission phase level and instrument level. At mission level, archive as root(isda) is defined. Under archive, mission wise bundle (ch2_bundle) is created. In this mission, there are three sub bundles for orbiter, lander and rover is created. Under mission wise sub bundle different mission phases are defined viz earth bound phase(ebp) and normal operation phase (nop). Under each mission phase level, collection of different instruments is defined. Below Figure 1.1 shows mission level bundle archive structure.

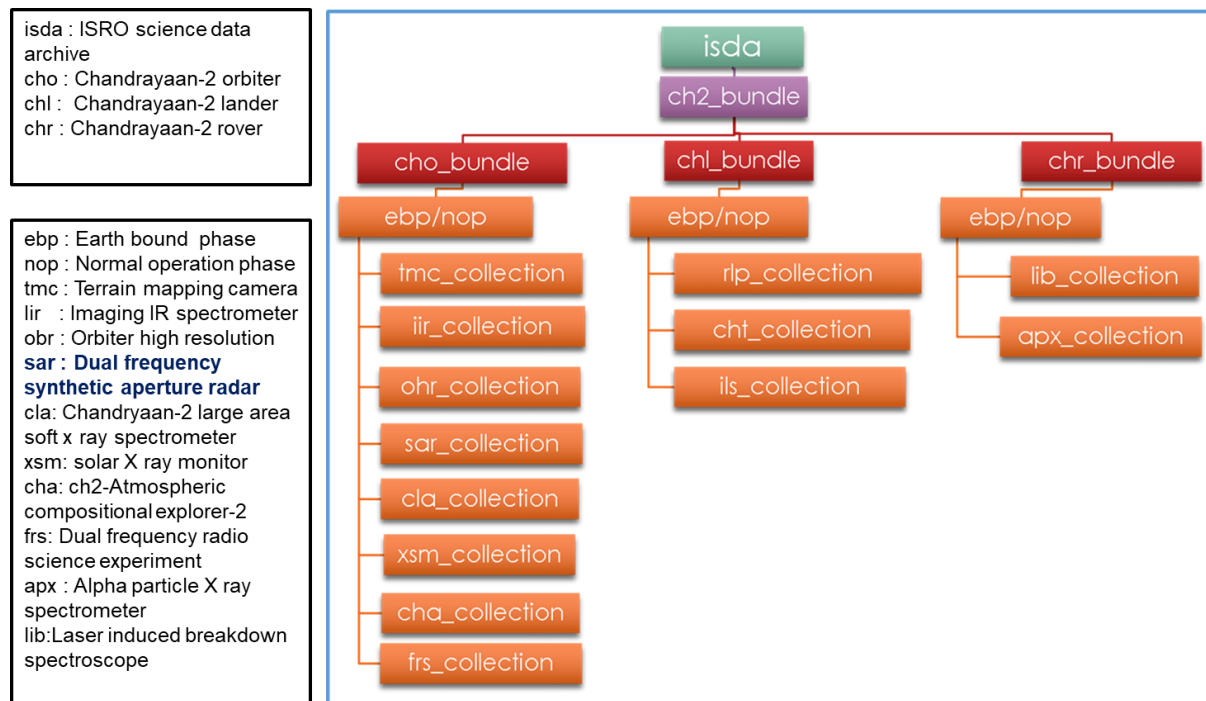


Figure 2.1 Mission level bundle archive structure

Next level instrument wise collection is defined. Under each instrument wise data collection, data is organized under each directory defined by PDS4. Data is organized based on the levels of data processing and data products definitions. All the data is plan to store as year, month and date wise. Below figure shows detail instruments wise collection structure.

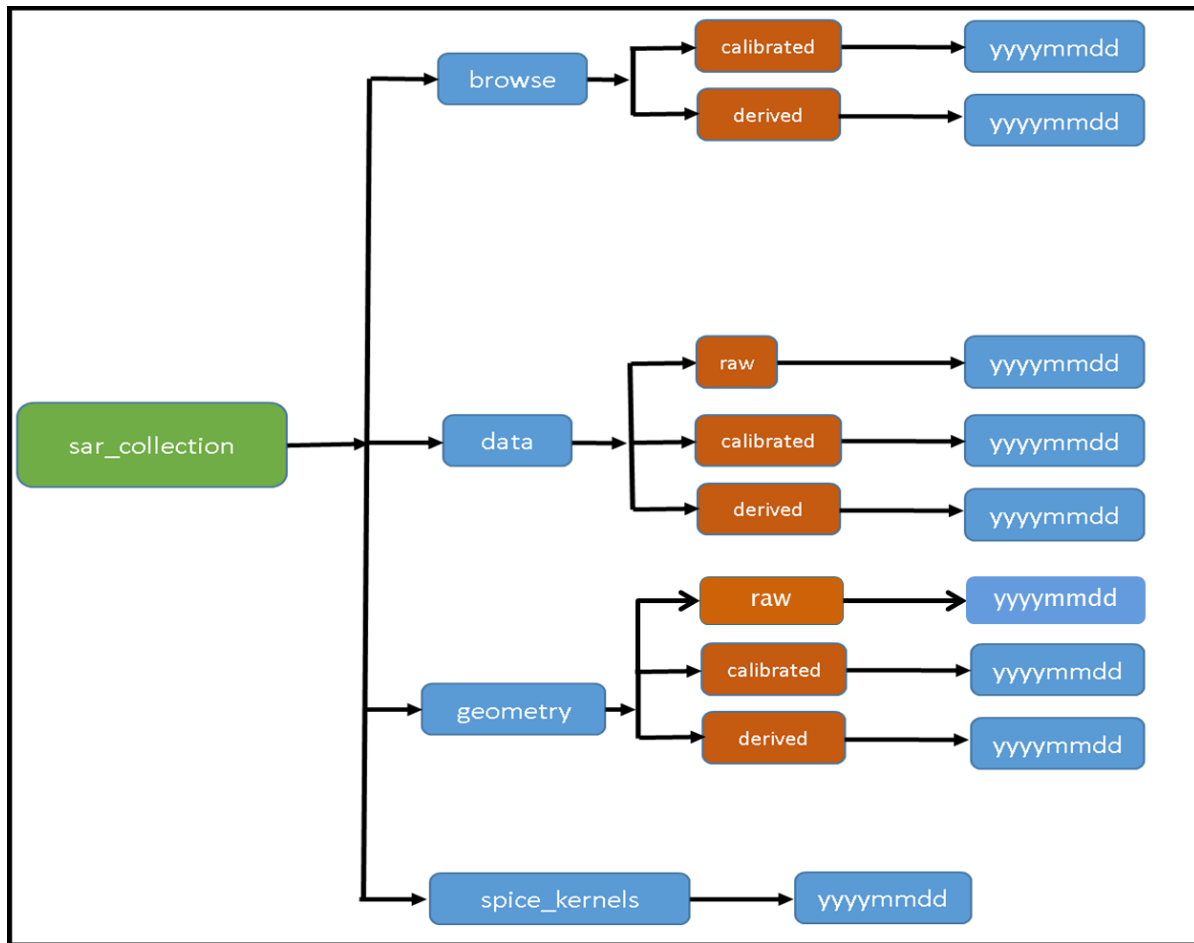


Figure 2.2 Instrument wise collection structure

2.1. Archived Data Products

A PDS product consists of one or more digital and/or non-digital objects, and an accompanying PDS label file. Labelled digital objects are data products (i.e. electronically stored files). Labelled non-digital objects are physical and conceptual entities which have been described by a PDS label. PDS labels provide identification and description information for labelled objects. The PDS label defines a Logical Identifier (LID) by which any PDS labelled product is referenced throughout the system. In PDS4 labels are XML formatted ASCII files. The PDS4 Data Products for SAR defined based on the defined data processing levels (L0, L1, L2 and L3) as Raw, Calibrated and Derived data products.

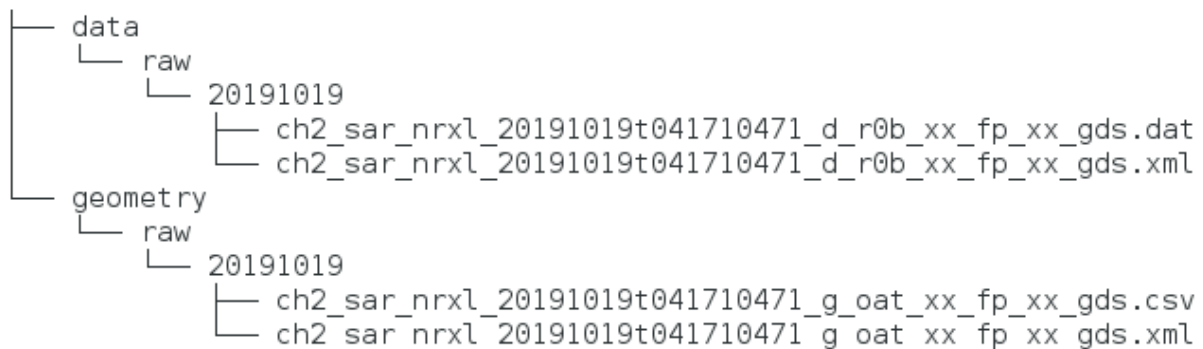
2.1.1 Raw data

The raw data records (RDRs) are in the form of raw binary files which have been extracted from frames downlinked from the instrument. The science data will be BAQ uncompressed. Then the auxiliary data along with science data i.e. complex in-phase and quadrature signal (I and Q format) is reassembled into contiguous radar range lines. Time information per PRF is also provided. There is a fixed size scan header in the beginning of each frame. Data has been corrected for missing frames by filling with zero. For more information, kindly refer to CH2DFSAR DPSIS document [1].

Table 2.1 Level 0 Data product details

Product	File type	File name extension	Description
Raw packets	Generic binary	dat	Raw binary file with time, payload auxiliary information and Science data per PRF.
Raw SAR Label file	Formatted text	xml	The raw SAR label file
Raw OAT	CSV	csv	It contains orbit and attitude information
Raw OAT Label file	Text	xml	Raw OAT label file

Example Level0 data product directory tree structure is as shown below

**Figure 2.3** Level0 data product directory tree structure

2.1.2 Calibrated Data Products

- **Level-1A** product consists of Slant range Image (SLI) in single look complex (SLC) format along with the label file in xml. The image is in TIFF format, and each individual polarization (RH/RV/LH/LV/HH/HV/VH/VV) has one TIFF file. These will be archived in calibrated directory under data directory. The associated grid file will be archived in calibrated directory under geometry directory.

Table 1.2.2 Level 1A SLC data products details

Product	File type	File name extension	Description
Image	Binary	Tif	The slant range complex image file. Each pixel is represented by two 4-byte floating point value (one 4-byte floating point real and one 4-byte floating point imaginary value). Each pixel in the slant range image is Seleno-tagged with a lat./lon. value.
Image File Label	Formatted text	Xml	This label contains information about the image.
Grid	Delimited Text (csv)	Txt	It contains the lat./lon. information corresponding to each pixel of the image. Generally, it is kept at a particular sampling in both azimuth and range direction. It also contains the Slant Range and incidence angle at the target point.
Grid file	Formatted	Xml	This label contains information about the grid.

Label	ed text		
OAT	CSV	Csv	It contains orbit attitude information
OAT Label file	Text	Xml	OAT label file

- **Level-1B** product consists of Ground Range Image (GRI) along with the Label file in xml format. The image is in TIFF format, and each TIF corresponds to individual polarization in case of dual/circular/Full polarization. These will be archived in calibrated directory under data directory. The associated grid file will be archived in calibrated directory under geometry directory.

Table 1.2.3 *Level 1B data products details*

Product	File type	File name extension	Description
Image	Binary	Tif	The ground range unsigned short int image file. Each pixel is represented by 2-byte unsigned short int. Each pixel in the slant range image is Seleno-tagged with a lat./lon. value.
Image File Label	Formatted text	Xml	This label contains information about the image.
Grid	Delimited Text (csv)	Txt	It contains the lat./lon. information corresponding to each pixel of the image. Generally, it is kept at a particular sampling in both azimuth and range direction. It also contains the Range and incidence angle at the target point.
Grid file Label	Formatted text	Xml	This label contains information about the grid.

- **Level-2** product consists of Map projected Seleno Referenced Image(SRI) along with the Label file in xml format. These will be archived in calibrated directory under data directory. The associated grid file will be archived in calibrated directory under geometry directory.

Table 1.2.4 *Level 2 data products details*

Product	File type	File name extension	Description
Image	Binary	Tif	The Map projected image file. Each pixel is represented by 2-byte unsigned short int file(amplitude).
Image File Label	Formatted text	Xml	This label contains information about the image.
Grid	Delimited Text (csv)	Txt	Each pixel in the image is Seleno-referenced with a lat./lon. value. Here lat./lon. value represents the image pixel values with reference to true north.
Grid file Label	Formatted text	Xml	This label contains information about the grid.

Tree structure of calibrated Full Polarization data is shown below:

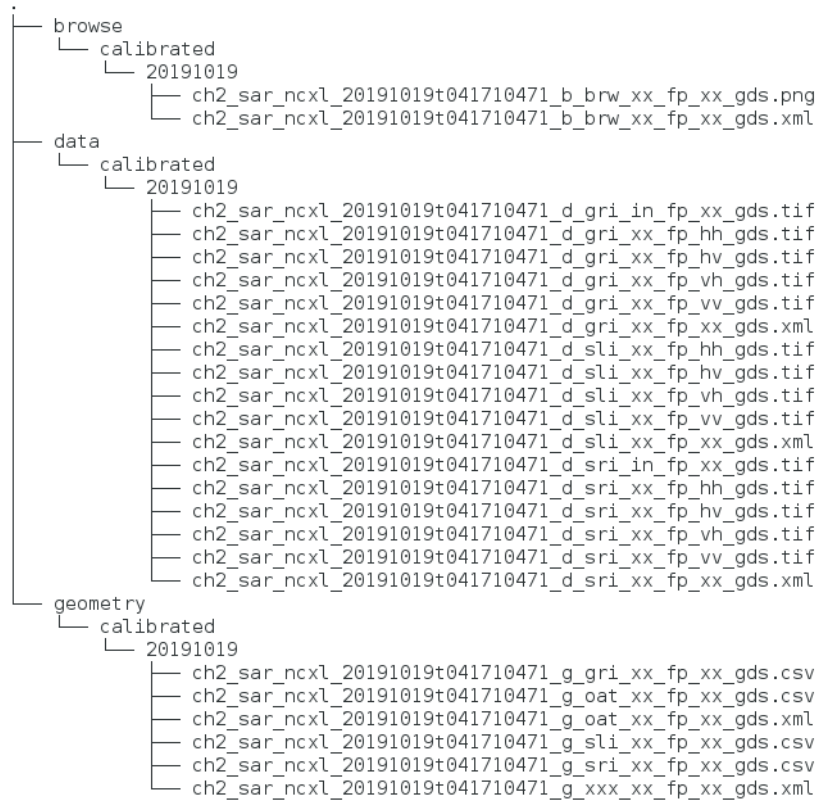


Figure 2.4 Full Pol data product directory tree structure

Similarly, tree structure of calibrated Circular Polarization data is shown below:

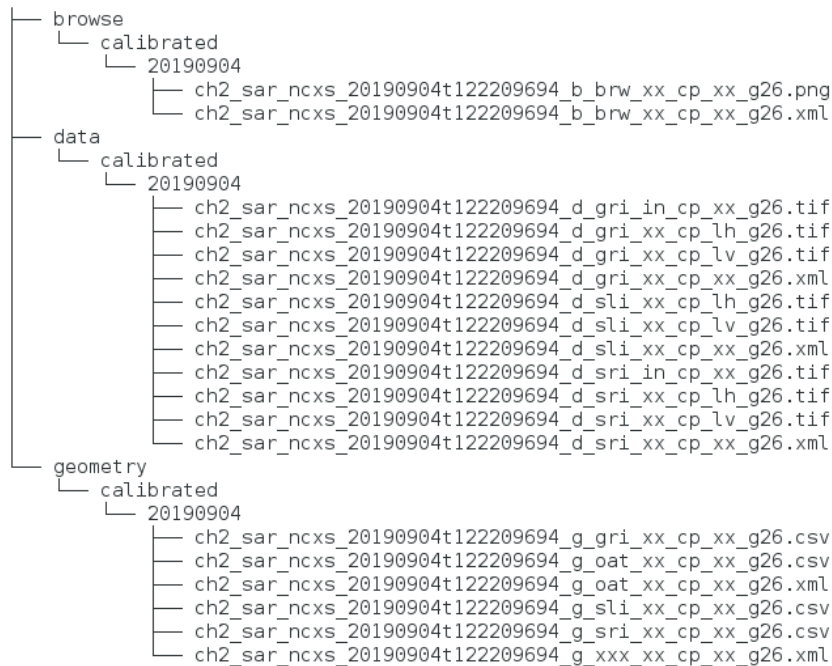


Figure 2.5 Circular Pol data product directory tree structure

3. Tools

Basic tools for opening data products and there datatypes are described in below table

Table 3.1 CH2DFSAR data products

Product	Type	Datatype	Tools
Level-0(Generic Binary)	dat	Signed Byte	Any generic binary Image viewer
Level-1A sli (single look complex image)	tiff	ComplexLSB8(I/Q interleaved data with 4 bytes each)	Any Tiff file Image viewer like QGIS,SNAP, Midas etc.
Level-1B gri (ground reference image)	tiff	Unsigned short int	
Level-2A sri (Seleno referenced image)	tiff	Unsigned short int	

4. Conclusion

This user guide is abstract view of PDS4 Data Products. DFSAR image acquisition started from September 2019. DFSAR is operated in Dawn-Dusk orbit continuously for three months. The data downloaded from the spacecraft is processed in ISSDC and PDS4 data products were generated and archived under active archive structure. At regular interval, Long Term Archive (LTA) was prepared and distributed to users from Chandrayaan-2 Map based browse application hosted in ISSDC, ISTRAC. For more information regarding product like file naming convention, data format, Level of products etc, kindly refer to CH2DFSAR DPSIS document [1].

5. Abbreviations

Table 5.1 provides the list of definitions, abbreviations and acronyms.

Table 5.1 List of definitions, abbreviations and acronyms

S.No.	Abbreviation/Term	Full form/Definition
1	ISSDC	ISRO Space Science Data Center
2	SLC	Single Look Complex Image
3	GRI	Ground Reference Image
4	SRI	Seleno Reference Image
5	PDS	Planetary Data System
6	ebp	Earth Bound Phase
7	nop	Normal Operation Phase
8	DFSAR	Dual Frequency Synthetic Aperture Radar
9	ISTRAC	ISRO Telemetry Tracking and Command Network

10	MDPD	Microwave Data Processing Division
11	SIPG	Signal and Image Processing Group
12	LTA	Long Term Archive
13	XML	eXtensible Markup Language
14	OAT	Orbit Attitude

References

- [1] Data Product Software Interface Specification and PDS-4 Archival Chandrayaan-2 Dual frequency SAR SAC/SIPG/MDPD/CH2/SAR/2018/04/03/v1.1
- [2] L- and S-band Polarimetric Synthetic Aperture Radar on Chandrayaan-2 mission , Space Applications Centre, Ahmedabad 380 015, India , CURRENT SCIENCE VOL. 118, NO. 2, 25 JANUARY 2020 ,doi: 10.18520/cs/v118/i2/226-233